Visual display of qualitative information: Implications of using illustrations to depict video data

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[This paper is part of the Focused Collection on Qualitative Methods in PER: A Critical Examination.] Qualitative studies in the domain of physics education research have become more common in the last several decades. Methodologically, this has been marked by an expansion of the types of data collected in physics education research (PER): namely, in the use of individual and group interviews, problem-solving sessions, and classroom observations—often in the form of video data. However, depictions of those video data in publications, a key opportunity for communicating analytic approaches and results, are often left out or are constructed without addressing some important implications of video data depictions. In this paper, we critically examine how video data can be depicted, first providing a rough taxonomy of depiction approaches. We then emphasize the case of line illustrations as a depiction approach with significant potential for improving (i) the transparency of analytic perspectives, (ii) communicational clarity, and (iii) the handling of ethical issues, such as participant anonymization, followed by a discussion of the limitations and challenges of using illustrations. In the Appendix, we provide a practical guide for researchers interested in making their own illustrations for academic publishing purposes.

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I. INTRODUCTION

Conducting physics education research (PER) entails processing and representing data—a point that holds regardless of whether those data are quantitative, like numbers in a table, or qualitative, like videos of human interactions. However, while the practice of processing and displaying quantitative information has received a fair amount of attention and discussion [1], the processing and displaying of qualitative data is a practice that has largely gone undiscussed within the PER literature. Nowhere is this clearer, perhaps, than with how physics education researchers choose to depict video data in their publications.

Video data have become a recent staple of qualitative PER work, especially as an increasing number of physics education researchers have begun to value the nonverbal components of communication such as gesture, gaze, manipulation of objects, and body posture as noteworthy modalities in the teaching and learning of physics [2–9]. The ways in which physics education researchers have depicted their video data in publications vary greatly, ranging from text-only descriptions to unprocessed frames of video. Yet, a discussion about the implications of different approaches to video data depiction is often absent. Such a discussion can address issues such as how best to convey the analytic perspective of the researcher, how to communicate clearly and accessibly to a diverse range of readers, and how best to deal with ethical questions, such as the anonymity of research participants. With these factors and more in mind, we argue that one promising approach to representing video data is through the use of *line illustrations*.

We ground our discussion in the ongoing academic conversation regarding the handling of video data, examining how scholars doing qualitative work have grappled with the demands of depicting video data [10,11]. We then highlight examples of video data representations in PER and show what line illustrations stand to afford the interdisciplinary, multiprofessional readership of journals like *Physical Review Physics Education Research*, as well as discuss the limitations and possible drawbacks of their use. For those interested in starting to use illustrations in their work, or to further develop their illustration technique, we also provide a separate practical guide as an appendix.

While this overview paper will discuss line illustrations as a particular approach to representing video data,

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ultimately our aim with this work is to highlight some worthwhile considerations for physics education researchers engaged in qualitative research and to provide a possible starting point from which to better grapple with the challenges of displaying qualitative data.

II. BACKGROUND: USING AND DEPICTING VIDEO DATA

A. A history of video as data: From psychology and anthropology to PER

Here we present one history of the use of video data in research, focusing specifically on interpretivist approaches to qualitative research with an emphasis on studying multiple modalities of human expression. Researchers have long been examining the embodied expressions of humans through direct observation and photographic records [12,13]. By the mid-20th century, however, developments in video recording technology enabled a new, systematic type of human interaction research, rendering a methodological frontier that initially attracted psychologists, psychiatrists, linguists, and anthropologists in interdisciplinary research groups like Stanford University's *Center for Advanced Study in the Behavioral Sciences* [14]. These early video data researchers—including such scholars as Ruesch [15] and Birdwhistell [16]—were all in some manner examining the contributions of nonverbal behavior alongside their verbal counterparts. They began developing a picture of human social interactions as rich *semiotic ecologies*, where each interaction was enacted in a manner that was "not only *multimodal* (i.e., involving nonverbal as well as verbal communicative means) but *multiparty* (i.e., involving listeners' activity in conjunction with that of speakers" [14] (p. 181). Systematic study of such semiotic ecologies was made possible through the continuous audiovisual accounts of video data.

For example, starting in the 1970s ethnographic anthropologists like Erickson and Shultz [17], McDermott *et al.* [18], and Mehan [19] offered some of the first published exhaustive observations of elementary school classrooms through their use of portable video cameras recording to tape. Goodwin [20] famously used video data to examine how the single phrase, "I gave up smoking cigarettes one week ago today actually," was augmented by a speaker's gaze during a dinner conversation. Spurred on by the findings of these early video research projects and the capabilities of video technology, *embodiment* soon began to garner attention in many academic disciplines beyond sociology or anthropology. This shift toward an interest in the body was described by Nevile [21] as the "embodied turn" of research on language and social interaction.

From the 1990s onward, researchers came to discuss how conversing individuals made use of their surroundings and their bodies in a mutually elaborating, environmentally coupled manner [22–24]. In education research, scholars have shown, for example, how gestures can aid physics students in conveying accurate ideas [25] and even facilitate idea construction by allowing speakers to "offload" their thinking [26,27]. Again, the insights coming from these studies were made possible to a large degree by meticulous analysis of video data.

Similar analyses of video data made their way into the PER community within the first decade of the 21st century [5]. Subsequently, PER has seen an influx of research that takes inspiration from the aforementioned work as well as other efforts in anthropology and education research. A growing number of PER scholars now use video as a central and often main source of data. In the last decade or so, methods taking further inspiration from multimodal conversation analysis and multimodal discourse analysis have been combined with other theoretical perspectives [2–4,8,28,29], increasingly featuring in *Physical Review Physics Education Research* as well as other PER and science education journals (e.g., *European Journal of Physics*, *Journal of the Learning Sciences*, and *Science Education*). In their review of PER as a field, Russ and Odden [30] highlighted that some of these publications stand to play an important role in pushing forward the development of qualitative methods in PER. The present paper aims to contribute especially to the efforts along this qualitative methodological frontier.

B. Depicting video data

To facilitate our discussion on the usefulness of illustrations for analyzing video data in PER, we first provide an overview of the approaches taken to depict video data in publications that focus on human interactions in the teaching and learning of science. We have collected examples from three scientific journals—namely, *Physical Review Physics Education Research*, the *Journal of the Learning Sciences*, and the *European Journal of Physics*—and organized them by some of the salient differences in how they depict video data (Figs. 1 and 2).

A first noticeable difference in approach is whether a single image is used (nonsequential) or if multiple images are used in sequence to convey some time evolution (sequential). For the sake of clarity, we have split our collected video data depictions into Figs. 1 and 2 by this first feature, respectively. Within these larger categories of nonsequential or sequential depictions, we further distinguish our collected examples according to four subcategories: (i) wide frames, (ii) narrow frames, (iii) processed frames, and (iv) illustrations. Wide frames typically comprise depictions of participants' full upper bodies, while narrow frames are generally examples of cropping a frame of video down around a specific portion of a scene (e.g., only the hands of a participant or only a single participant among a full group of recorded participants). Next,

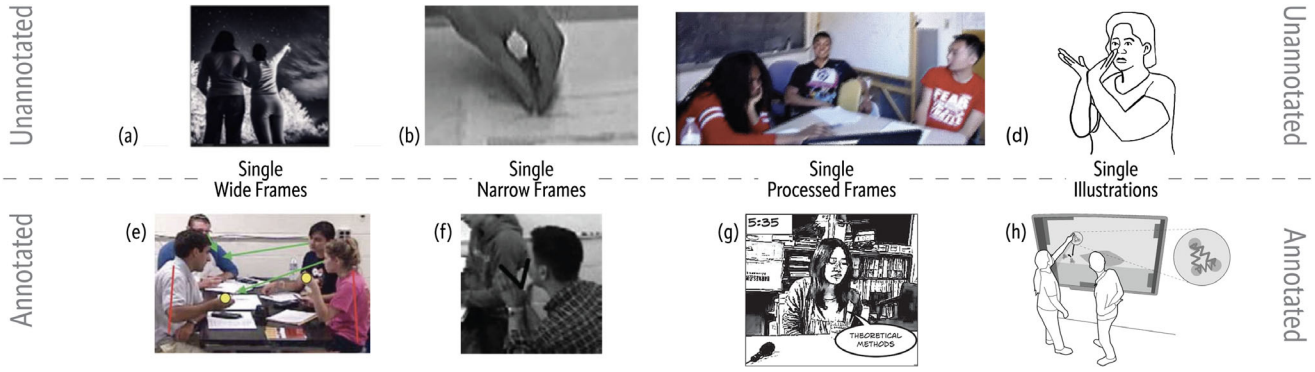

FIG. 1. Examples of nonsequential depictions of video data, categorized as wide frames, narrow frames, processed frames, and illustrations—all with unannotated and annotated variants. (a) is an AI-generated image resembling the figures in Azevedo and Mann [31] (for copyright purposes). The remaining images are reproduced under a Creative Commons license from the following sources: (b) Hu and Rebello [32], (c) Sabo and Elby [33], (d) redrawn in the style of Alibali and Nathan [34] (for copyright purposes), (e) Frank and Scherr [35], (f) Scherr [5], (g) Gupta *et al.* [36], and (h) Euler *et al.* [37].

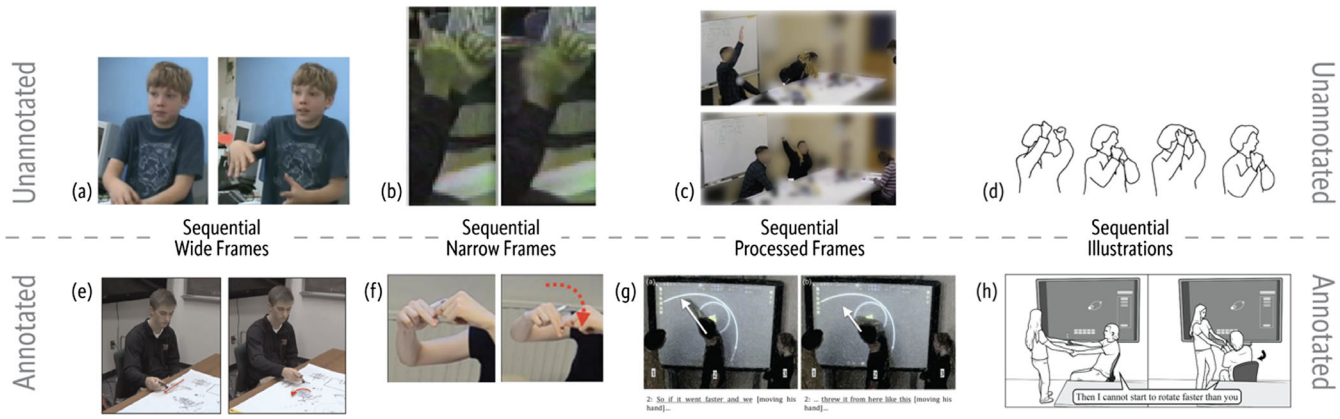

FIG. 2. Examples of sequential depictions of video data, otherwise following the same categorization as Fig. 3. Images are reproduced under the Creative Commons license from the following sources: (a) Phillips *et al.* [38], (b) Stephens and Clement [39], (c) Robertson and Hairston [28], (d) Stephens and Clement [39], (e) Odden [4], (f) Eriksson *et al.* [40], (g) Gregorcic *et al.* [3], and (h) Euler *et al.* [2].

processed frames are blurred or filtered in some way that alters the original image captured by the camera. Finally, illustrations refer to line drawings (typically, though not exclusively, black-and-white)—ostensibly made by tracing frames of video in some manner. For each of the four subcategories, an unannotated and an annotated example is included. In this case, annotation means some type of arrows, speech bubbles, or insets are added on top of the rest of the image.

We stress here that our aim is not to provide an exhaustive account of the different depictions in existence or to determine quantitatively the frequency of their use. Rather, the aim of Figs. 1 and 2 is to roughly map the space of approaches to video data depiction and to use this as a starting point in the discussion of some considerations of depicting video data, with a special focus placed on illustrations. As these figures show, the range of depiction styles is quite broad. While a consensus on the best ways of depicting video data is not realistic to expect, or even desired, there is almost no discussion on this topic present in the papers from where these figures originate.

In Fig. 3, we have constructed new video data depictions that all stem from the same video frame—i.e., an unannotated, wide frame pulled from the video data uploaded as supplementary material of Scherr [5]. Only the single frames or illustrations are shown in Fig. 3 (no sequential frames), but having the same content presented across the various subcategories should allow for easier comparisons of what we mean by each category type. In the sections that follow, Figs. 1–3 are used as examples for discussing the implications of illustrations for depicting video data.

III. THE IMPLICATIONS OF ILLUSTRATIONS

Having provided an initial map for the space of video data depictions, we now focus our attention on the factors to consider when depicting video data in qualitative PER

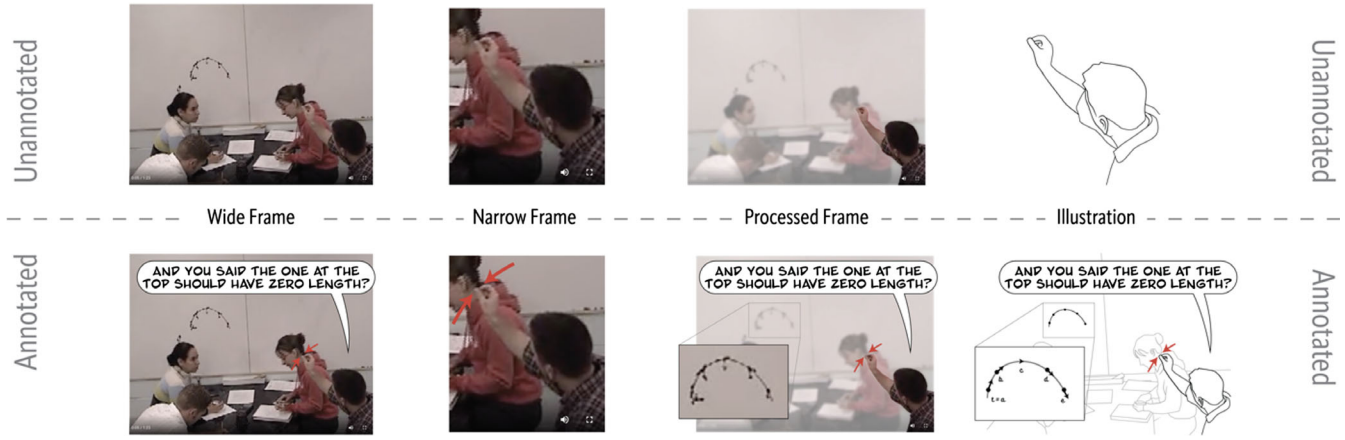

FIG. 3. Examples of the same source frame of video transformed into the various types of video data depictions, following the same categorization scheme as Figs. 1 and 2. The original frame of the video (upper left) was sourced from the supplementary videos of Scherr [5], while the remaining depictions were generated by us for the purposes of this paper.

work through the case of line illustrations.[1] This is because, especially when compared to the other approaches for depicting video data, we argue that illustrations have important implications for analytical, communicational, and ethical considerations of qualitative PER work.

A. Portraying analytic perspectives

Hammer and Berland [10], building on the arguments of Hall [41], emphasize that video data are inherently theory laden. That is, the analytic perspectives of a researcher are necessarily influential at virtually every stage of video data collection: the room(s) into which a camera is brought depends on the analytic perspective of the researcher, the framing of the camera shot depends on the analytic perspective of the researcher, etc. This not only results in a blurry line between what we should consider data and what we consider claims about that data [10] but also serves to undermine the illusion that video data are objective or theory neutral.[2] Bezemer and Mavers [11] emphasize the fact that depictions of video data are laden with theory as well. However, in moving beyond Hammer and Berland's [10] cautionary note about the nature of video data, we argue that the theory-ladenness of video data depictions can actually be an advantage for communicating researchers' analytic perspectives. Through processes we refer to as highlighting and supplementing, illustrations can convey a researcher's analytic perspective by visibly encoding the decisions taken during their analysis of video data and consequently making them available for academic scrutiny.

One way illustrations portray the researchers' analytic perspective is through so-called highlighting—i.e., a process by which the reader's attention is drawn to "elements of the focal interaction [through the] foregrounding of certain features" [11] (p. 195). All depictions of video data entail highlighting to an extent, such as through the selection of moments deemed critical and the omission of moments deemed irrelevant. However, illustrations especially emphasize the details attended to by the researchers in their analyses, bringing them forth explicitly for the reader by virtue of the simplification inherent to the depiction style. Importantly, the process of highlighting in video data analyses is guided by the researchers' analytical purposes [11], so for the reader, the highlighting decisions apparent in the video data depictions become a window into the analytic perspectives taken in the publication. Researchers using illustrations can literally outline the features of the video data that were germane to their analysis, including what they deemed relevant and excluding what they deemed irrelevant.[3]

Compare this with screenshots or frames of video that are often used in publications. With single frames of video like all the examples in Figs. 1 and 3, authors tend to convey which instances in time among their video data they

[1]It is important to note that the illustrations we discuss throughout this paper are manual *tracings* by the author(s) over frames of video data (see the Appendix for details). We have occasionally encountered the suggestion to use automatic "photo-to-sketch" tools as a shortcut for the task of tracing manually. This is likely what was used for the figures in Ref. [36] [shown in Fig. 1(g)]. However, it is less clear if and when such automatic approaches successfully portray the analytic perspectives of the researchers (Sec. III A) and address challenges regarding participant ethics and anonymization (Sec. III C).

[2]A point often alluded to when comparing video data to other qualitative data sources such as field notes or observer accounts [42].

[3]An analogous example from physics helps us illustrate an important point here. Circuit diagrams are still generally preferred to photos of circuits and digitally rendered schematic diagrams are often preferrable to photos when conveying the intricacies of a complex experimental setup. Outside of physics, it is still common for biological reference manuals to include illustrations because of the effective ways in which illustrations can convey the most relevant aspects of a specimen [43].

judged as relevant, but often only weakly highlight which details within those images were explicitly attended to during analysis. Some researchers opt to considerably crop images to delimit the relevant body parts tracked in their analysis (e.g., the narrow frames in Figs. 1 and 2). However, doing so often comes at the cost of decontextualizing the movements of those body parts with respect to environmental features and/or with respect to the rest of participants' body. This can be seen in Fig. 3 where the narrow frames of the second column both highlight the single participant with their arm outstretched but do so at the cost of cropping out the drawing on the whiteboard to which the participant is gesturing in an environmentally coupled manner [24].

By contrast, illustrations like the one in the bottom right of Fig. 3 can portray moments of video data to the reader in a form that is holistic in setting, yet selective in the level of detail presented. Note how, especially compared to the wide, unannotated frame in Fig. 3 (top left), the annotated illustration of Fig. 3 more clearly depicts the aspects of the frame to which the researcher attended through the use of the darker outline of the gesturing participant and the enlarging of the whiteboard detail (as per the discussion of supplementing below). With illustrations (and processed, annotated frames like the one in Fig. 3), subjects can be presented as clearly outlined individuals elaborating on meanings in the context of the relevant environmental features. Depictions such as these thus have the potential to more clearly communicate the so-called "professional vision" [11,22] of the researcher, in many cases better than text-based transcripts accompanied by frames of video.

Second, while highlighting within an illustration can be thought of as the filtering out of details deemed less relevant by the researcher, supplementing within an illustration is an additive layering of details to extend qualitative data beyond the discernible details of the filmed interaction. Researchers often collect additional data about the scene beyond the details captured by a video camera sensor. For example, a researcher who was present in the room may have been able to clearly see the details of what participants were writing on a whiteboard, yet, at the final resolution of the video data those details of the participants' writing might end up being illegible. If the participants' writing is relevant to the researchers' analysis of the participants' interaction, then, it is incumbent on the researchers to supplement their depictions of video data with those written details as they saw them in person (see the inset of the whiteboard drawing in the bottom right of Fig. 3).

In Fig. 4, an illustration of student interaction in front of an interactive whiteboard is supplemented by insetting a closeup of the digital menu around which the students had been gesturing [37]. The inset provides a magnified view of the menu that is obscured by the student's body in the original video frame. The magnified portion of the scene is not only enlarged but also rotated toward the reader so that

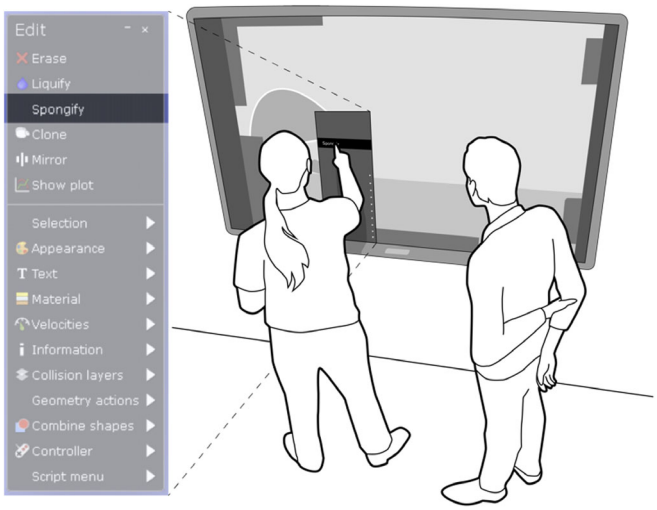

FIG. 4. An example of supplementing an illustration of video data with a closeup of a digital menu and a student's environmentally coupled gesture. Reproduced from Euler *et al.* [37] under the Creative Commons license.

the labels of the menu can be more easily read. By supplementing an illustration with screenshots of a digital display like this, the depiction of the video data can incorporate aspects of the environmental (in this example, technological) artifacts of the scene that often provide crucial context for the actions and utterances of the participants as they interact.

Another source for supplementing video data depictions can come not from extra details from the interaction that the researcher had access to beyond the video footage, but from the context that the researcher had from watching the video "play" through time. For example, during the act of watching an advancing video clip, the path traced out by the arc of a participant's gesture would be obvious to the researcher(s). When depicting this event in a static image during publication, researchers often portray this motion through some type of supplemented depiction—e.g., by annotating an arrow on top of the scene and/or by compositing an "action sequence" rendition of the motion that superimposes several locations of the participant's arm throughout the gesture (as is typical of sports photography). Again, illustrations make such an approach especially achievable. With illustrations, the researcher has the representational control to duplicate only the desired moving element of the scene (e.g., a moving arrow on a screen or a moving arm in a gesture; Fig. 5). This may mean that other background elements of the scene are rendered artificially static by comparison, but we emphasize that this pseudo-multiple-exposure technique should be viewed again as an opportunity for the researcher to portray their analytic perspective rather than reproduce some "objective" reality.

A final point is warranted relating to the theory-ladenness of video data and their depictions. Conspicuously constructed depictions of video data like illustrations have

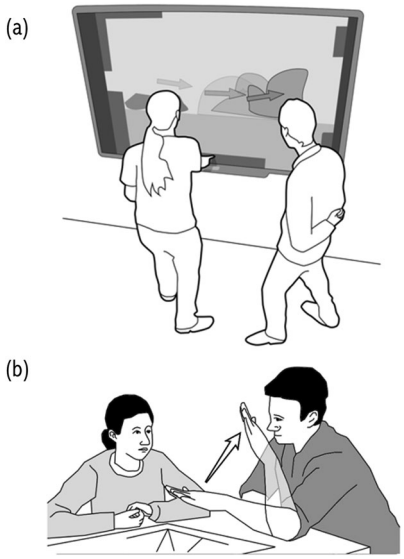

FIG. 5. Examples of single illustrations (nonsequential) with multiple-exposure-type "action sequences" for depicting the motion of (a) environmental elements (in the simulation, an arrow projectile flies from left to right and gets jammed in a soft object) and (b) gestures. (a) is reproduced from Euler *et al.* [37] under the Creative Commons license and (b) was redrawn in the style of Alibali and Nathan [34] (for copyright purposes).

metarepresentational markers that convey the degree of judgment inherent to video data analyses. That is, by virtue of being obviously researcher-made, illustrations frame the enterprise of analyzing video data as a subjective process of the researchers' making. Bezemer and Mavers [11] emphasize that the features of video data depictions themselves communicate to the reader how "real" something is: "image, like writing, can be used to suggest […] whether [a statement] should be read as an abstraction, or as a concrete instance, as imagined or real objects and processes" (p. 201; see also, [44,45]). It follows, then, that illustrations of video data—along with other obviously processed, annotated depictions—carry with them the implicit message that analyses necessarily rely on the perspective(s) taken by researchers. Conversely, attempts to preserve the "reality" of video data by depicting them with unadulterated frames or as video clips at least partially sustain the compelling, yet mistaken illusion that video data are objective [10,41].

B. Clarity of communication

Effective use of figures can be critical for communicating science to other scientists and to the general public [46], a point which stands to reason for the context of video data as well. Perhaps counterintuitively, when researchers are making a specific point about a time-evolving system, a static image is often better to support that point than a video or an animation: as Tversky [47] explains, animated depictions "are all too frequently too complex to be adequately perceived" (p. 38). Yet, as Jambor *et al.* [48] note, "figures in papers outside of one's immediate area of expertise are often difficult to interpret, marking a missed opportunity to make the research accessible to a wide audience" (p. 2). This issue of figure clarity is especially important in fields like science education research and PER, where the readership includes a wide variety of educators and researchers with a broad spectrum of research experiences. The clarity of figures can also have major repercussions for visually-impaired and low-vision readers. Thus, figures in publications, including the depictions of video data, should be made maximally clear as part of communicating effectively regardless of readers' sensory abilities (i.e., in accordance with the principles of universal design [49]).

On this front, illustrations can simply be crafted to be clearer for a reader than most frames of video. While a frame from a video recording may be easily interpretable by the researcher who was present in the room during filming or who has watched the video several times during analysis, it is certainly not the case that a newcomer to the data will consistently perceive the same level of detail (i.e., to distinguish the "signal from noise") in an image they are seeing for the first time. This is especially true when video data are collected in authentic teaching environments with nonideal lighting [31] and/or with backgrounds filled with elements other than the researchers' focus. These naturalistic video data and the frames selected from them can end up being low-contrast, low-resolution, and visually cluttered. Consequently, it is often more difficult for a reader to glean the relevant information from such representations. An illustration, on the other hand, can be crafted as a high-resolution, high-contrast depiction that omits the competing features of an interaction the reader need not focus on for the purpose of following the argumentation at hand.

For example, in the narrow, black-and-white frame of the video shown in Fig. 1(f), the authors depicted a participant creating an upward-facing "v" shape with their hands, emphasized by a thick black line annotation [5]. This line annotation is ostensibly quite necessary for picking out the participant's gesture against the competing background of similar gray values, yet even the annotation (which obscures much of the hand shape it was meant to highlight) reads relatively poorly within the dark grays of the video frame. Compare this with a similar gesture depicted in Fig. 1(d) [34]. Here, the authors portray the participant as isolated from any competing background elements and the v gesture is clearly visible from the high-contrast outlines. The latter depiction, an illustration, is noticeably clearer. A similar comparison of clarity can be made between the unannotated, narrow frame, and the unannotated illustration in Fig. 3—though the color version of the narrow frame already improves the readability of the frame of the video by providing another channel of contrast between the participant's gesture and the background elements.

The high contrast of illustrations also means that they remain readable at a wider range of scales. This is especially applicable for PER journals such as *Physical Review Physics Education Research*, where figures are often required to be downscaled to fit within the width of a single column. Vector-based illustrations (see the Appendix) do not take on any sort of defined resolution until they have been converted to a raster version for printing or embedding in documents. Thus, especially when illustrations are created as vector-based graphics, the scalability of video data depictions can be maximized. Vector-based graphics can also transfer effectively between "light mode" (dark text on bright background—a positive contrast modality) and "dark mode" (bright text on dark background—a negative contrast modality), which is often used to reduce eye strain when using a computer screen.[4] These features can make a big difference in improving the accessibility of images for diverse groups of readers.

Another approach taken by many scholars for depicting video data in publications, exemplified by the area of conversation analysis [50,51], is to portray the particularities of video data through highly detailed, text-based transcripts. To capture the nuances of human speech and interaction, such transcripts will often involve entire systems of symbolic transcription conventions including various brackets, dots, time markers for the lengths of pauses, and more [52]. Such comprehensive transcription approaches do well to crystallize the analytic focus of the researcher, yet they often require expertise with the transcription conventions for a reader to readily discern the nuances of the researcher's focus from the text alone. Therefore, if such transcripts are used in a publication, it can be advantageous to provide maximally clear depictions of the video data scenes alongside symbol-heavy text transcripts. As noted above, the depictions can then function as an entry point for nonspecialists to build up a familiarity with conventions by leveraging their visual acuity for reading human interactions.

Alternatively, when the aim of the transcripts is not to facilitate a detailed analysis of utterances in the style of conversation analysis, a more accessible representation of participant dialogue may be in order. Speech bubbles, which can be meaningfully integrated with illustrations (in the familiar form of a comic strip; see bottom half of Fig. 3, for example) present the video data in a way that is easily interpretable to a broad audience, because of the widely understood conventions for representing verbal and nonverbal information alongside each other in comics. That being said, full transcripts may still be of interest to a subset of readers and can be included with publications as supplementary material (see Refs. [2,9], for example).

C. Participant ethics and anonymization

Many of the points raised in this paper point to the utility of illustrations for qualitative research with a pronounced embodiment or interactionist angle. However, a consideration of ethics and anonymization suggests that illustrations could be useful for a broader audience of researchers using video data.

Representing video data in publication requires contending with the ethics of personal information and anonymity. Clark [53] asserts that it is "common practice […] to protect the identity of those who participate in research" (p. 4). However, when it comes to depicting video data in publications, PER scholars often make choices regarding what participant information to share and what to protect. For example, some use pseudonyms to protect the names of their participants, while depicting those same participants' faces in frames of video with no alteration (e.g., [4,54–56]). This suggests that participants' names and other written personal identifiers (such as student ID) numbers are more likely perceived by researchers as paramount to protect because such identifiers are searchable in databases, while faces are not. However, though it may have been true historically that faces were not searchable in the same way that names are, the public searchability of faces is becoming more of a reality as a consequence of facial recognition algorithms backed by machine learning.[5] Thus, the anonymization attempted by the use of pseudonyms may be compromised by unaltered frames of participants' faces. Although the risks associated with most education research are often considered minimal—especially when juxtaposed against the risks associated with other human subject research areas like medicine—these risks are rarely truly zero. Examples, where students' anonymity is especially important, include research on student learning difficulties or other cases where a normative judgment on individuals or groups of participants' academic performance or expression of opinion is likely to be made by readers. It is also important to protect the anonymity of underrepresented groups of students, who could, because of often very low representation in physics departments, potentially be identified by others. This is especially important when they share their experience, perhaps negatively reflecting on teachers or other students, which could put them at risk of retaliation.

The proposed default position in video-data-based publications is thus that participants' identifiable information, including their faces, be anonymized. It is incumbent on the researcher to only reveal the identities of their participants

[4]Most journals still rasterize figures during publication, which nullifies this light mode or dark mode capability. Especially now that most academic publications are read through screens, a move toward vectorial graphics is more justifiable than ever.

[5]Such as the controversial *PimEyes* tool, at the time of writing.

FIG. 6. Hybrid video data depictions reproduced under the Creative Commons license from Asplund and Kilbrink [57], blending elements of video frames, and illustrations.

when it is necessary for the communication of the results and after a careful risk-benefit analysis.

Many papers make no effort to obscure the faces of participants, even when there is no apparent benefit for not doing so. When researchers do attempt to protect the identities of their participants, it often involves obscuring the faces of the participants via an overlaid shape or some form of blurring (e.g., [28]). This is ostensibly a motivation behind depicting video data with "processed frames" such as shown in the third columns of Figs. 1 and 2. While this often does well to protect the identities of the participants, approaches that entirely obscure participants' faces effectively eliminate the possibility of including the participants' gaze or facial expressions as meaningful contributions in the visual depiction.

While participants may give consent for the data to be used without anonymization, researchers are responsible for thoroughly informing them what this entails. The consequences of publishing nonanonymized data can be hard to predict. One example illustrating this concern is the practice of publishing research data under a Creative Commons license (also used in *Physical Review Physics Education Research*), which makes the data freely available for reproduction, editing, and sharing, even when the values, intentions, or aims of those doing so do not align with those of the researchers or research participants. Furthermore, students' participation in a study may be subject to different interpretations through time (as sensibilities change) and across cultures or countries. It is our responsibility as researchers to consider these issues and inform the participants of the risks of revealing their identities, keeping the well-being of the participants as an utmost priority. While the use of illustrations alone can often resolve many of these issues, illustrations can also be integrated into hybrid depiction styles (see Fig. 6) to mitigate the challenge of maintaining the richness of "raw" video data frames, while still protecting the anonymity of participants.

This brings us to a final key benefit of illustrations. With an illustration, a researcher can outline participants' faces to a level of detail that does not fully resemble the participants' likeness, or they can alter the illustrated faces to the point that the displayed faces do not resemble the real faces of the participants. This allows the researcher to still convey enough facial detail to portray things like gaze or expression while protecting participants' identities. Other details, such as clothing, hairstyle, body type, and skin tone can also be portrayed, allowing the researchers to capture the diversity of participants. For further discussion of this topic, see the following section.

For example, hybrid illustrations such as Fig. 6 allow for the inclusion of interactional detail such as gaze, while preserving the participants' anonymity. If a picture or frame of video had been used here instead, the visual details of the speaking participants' faces would have been necessarily obscured. The use of illustration here allows for the ethical inclusion of the full head and face of the participant while maintaining their anonymity.

It is possible that researchers avoid depicting video data in any form precisely as a way to preserve participant anonymity. For researchers reluctant to portray their participants in their publications, illustrations are an option that allows for anonymity to be upheld while providing the communicational benefits of figures [58].

IV. LIMITATIONS AND CHALLENGES OF USING ILLUSTRATIONS

Depicting participants using line illustrations means altering the participants' likeness in a process of deliberate simplification. This can result in certain aspects of the participants' identities, such as race, gender expression, political views, or religion (as expressed through clothing or jewelry, for example) to become obscured. While researchers may see this as a desired feature of using illustrations, it can also be seen as problematic. Indeed, ignoring or glossing over certain personal characteristics can perpetuate the often-wrongful view that they do not matter [59]. A compelling case for the importance of accurately collecting and depicting information about the participants' identities can be made from the point of view of achieving better visibility and representation of diversity in physics, honoring the participants' identities, as well as signaling to the research community that aspects of identity such as race, for example, are a legitimate locus for researchers' attention.

At the same time, there are important considerations that should accompany the depiction of characteristics such as race, ethnic background, religion, gender expression, political views, etc. By generating such depictions, one runs the risk of perpetuating stereotypes. A possible way to address this challenge is to consult with the participants and get their feedback on whether the created depictions

represent the aspects of their identity in a clear and respectful way [60,61].

There are also important challenges at the intersection of the just discussed issues with research ethics. For example, in some legal systems (e.g., European Union, where the General Data Protection Regulation, or GDPR, is used), there are special rules regulating the collection and processing of personal information. A special category of sensitive personal information includes participants' ethnic background, religious views, political views, information on their sexual life, and more). In some countries (e.g., Sweden), researchers may not collect or process sensitive personal information unless one can make a case in an application to a national ethics board, that among other things, collecting such data is necessary for the purposes of answering the proposed research questions. In the European Union, GDPR, which regulates the collection and processing of all personal information for research (and other) purposes, places heavy emphasis on the protection of individuals' personal information. In such a context, the use of illustrations is appealing precisely because obscuring personal information in publications is a regulatory requirement.

Illustrations could also not be the optimal approach to depicting video data. This may be the case where the amount of detail that would be required to meaningfully depict all the important features is very high or that the tools and skills at the disposal of the illustrator are not sufficient to express important nuances in the data (e.g., complex textures and colors). There may also be other aspects of video data depiction that researchers wish to prioritize, such as a sense of realism or preservation of the "raw" data for future analyses, or analyses by others using different theoretical lenses. In these cases, other approaches to representing and anonymizing video data, such as local blurring and annotating of videoframes with relevant information may be more appropriate. A hybrid approach to depicting video data is one way of mitigating some of the limitations of illustrations. For example, Vossoughi *et al.* [62] superimpose line illustrations over the original frames of the video, thereby highlighting the relevant details of the scene and boosting the clarity of images through improving contrast (Fig. 7). As discussed in the previous section, Asplund and Kilbrink [57] selectively illustrate only the faces of participants in their video data depictions as an effective approach to anonymization (Fig. 6).

To authors of academic texts, a real limitation of illustrations is also that they take some time and effort to make. Taking into account the considerations brought up above, the decision to invest time into making them will be done on the level of a particular team of authors and their perceived ability to produce illustrations. It is, however, easy to overestimate the time, effort, and talent required for creating functional illustrations. As we show in a separate Appendix to this paper, the process can be relatively simple, not exceedingly time consuming, and does not require a lot of artistic talent.

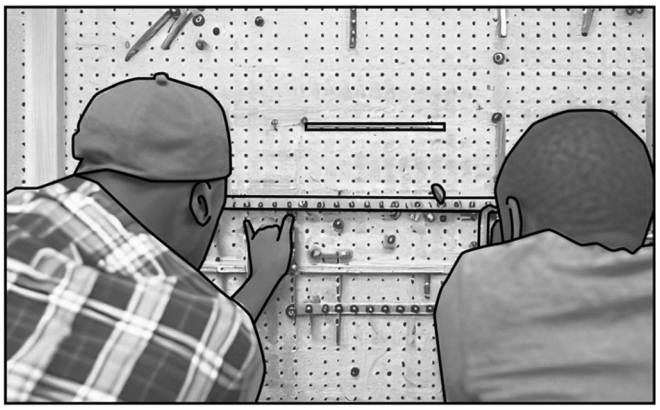

FIG. 7. Hybrid video data depictions in the style of Vossoughi *et al.* [62], blending elements of video frames, and illustrations. An AI-generated image was used for copyright purposes.

V. CONCLUSION

PER scholars examining video data of human interactions during the teaching and learning of physics can inform their approach to depicting video data through a consideration of the points raised in this paper, whether or not they use illustrations. Namely, researchers can consider (i) the extent to which a depiction portrays the researchers' analytic perspective to the reader, (ii) the clarity of a depiction for communicating to readers a range of sensory abilities or levels of expertise in qualitative research, and (iii) the efficacy of a depiction in conveying the desired level of visual detail while maintaining participant anonymity.

Illustrations can be used to clearly portray the theoretical perspective of the author and can be readily understandable to a broad audience. They are inherently flexible, allowing researchers to tailor the level of detail in the display of their data to the grain size of their analysis. Furthermore, they can allow for a nuanced handling of ethically challenging issues, such as the portrayal of participants' facial expressions while protecting their anonymity.

It is important to point out that the case we make in this paper for the use of illustrations in qualitative PER should not be undercut by the development of better cameras. The use of illustrations is not a step backward to prephotography techniques, but rather an intentional step forward (supported by modern digital tools) in representing video data, which can carry a different communicative power than photos or frames of video. In considering the potential that more PER scholars begin using illustrations to depict their video data, we acknowledge that for many researchers, this will entail learning a new set of technical skills (see the Appendix for practical advice regarding creating illustrations). However, it was not long ago that scientists were

expected to create diagrammatic illustrations for their publications by hand [63]. Indeed, there is a long legacy of illustration work that has been central to knowledge generation and communication for millennia [64]. PER scholars working with video data may benefit from incorporating this legacy with modern tools to better communicate their attention to the visual details of teaching and learning.

Whether or not researchers use illustrations in their depiction of video data, we outline some important considerations for dealing with and depicting video data in qualitative PER work. The PER community would benefit from more explicitly discussing the ways qualitative data are depicted in publications. We hope that this paper can offer a possible starting point for that conversation.

APPENDIX

Here we briefly describe the approach we have taken in creating illustrations for use in recent PER work. While this is certainly not an exhaustive discussion of the craft of digital illustration, we aim to provide an entry point for those getting started and demonstrate the relative ease with which high-quality versions of these illustrations can be made without previous expertise in graphic design or drawing.

1. General technique: Tracing over frames of video

Regardless of the approach taken to draw an illustration, we advocate for the general technique of taking frames of video and manually tracing directly over them in a digital drawing software. This first entails that a frame of video should be selected from video data at the desired point (a nontrivial selection process that is necessary for representing video data in static depictions, see Ref. [65]) and then captured as an image—which can be achieved through a "screenshot" of the media player. From there, the frame of the video should be imported into the drawing software of choice.

Drawing directly over a frame of video not only makes the task of drawing much easier than a freehand approach from reference but it also helps ensure that the final illustration closely matches the video data.

2. Drawing approaches: Stylus tracing vs Bézier curves

There are generally two approaches one can take to drawing the lines of an illustration. Both approaches depend on tracing over individual frames from video data to draw an outline around each relevant entity of the scene (e.g., the participants or objects in the environment), as well as a series of shorter detail lines within (e.g., for the participants' hair, clothes, or faces). The first approach, which we can call *stylus tracing*, entails drawing over frames of video with a tablet pen stylus (or similar) and resembles traditional tracing practices done with light tables and tracing paper. The second approach, *Bézier curves*, involves the piecewise drawing of lines through connected series of individually defined points, which one could imagine vaguely resembles a kind of "stitching" of the outline of the desired objects "puncture-by-puncture" with digital "thread."

The choice between these two approaches is often aligned to the software used to create the illustrations—of which there are many options available to the interested physics education researcher. This is to say, most (but not all) software is geared toward one or other of these illustration approaches, not both. We will discuss some viable software options for both approaches in turn, but it is also worth noting that publishable illustrations can be created with the traditional drafting tools of pencils or pens and paper. See Volkwyn *et al.* [66] for a recent example of where this "low-tech" approach was used in PER to good effect.

With either the stylus tracing or Bézier curves approach, it is critically useful to utilize so-called layers in the chosen illustration software. Layers allow for the isolation of the linework from the video frame source, akin to drawing on a separate tracing paper rather than directly onto an image itself. This is often not only necessary for the eventual export of linework (i.e., by hiding the source image layer during export) but can also allow the illustrating researcher to use techniques that improve readability while tracing (e.g., through the lowering of the source image's opacity).

a. Stylus tracing

Given some basic experience in drawing and access to a touchscreen with a stylus, the *stylus tracing* approach should be a good starting point for many education researchers. This approach entails following the outlines of relevant objects in a scene with continuous strokes, just as one does with a pen or pencil on paper. A main advantage of the stylus tracing approach is its shallower learning curve owing to most researchers' familiarity with using a writing implement. Usually, this approach will result in a rasterized (i.e., pixelated) outline that has a natural, hand-drawn aesthetic. However, some aspects of editing images are often harder with this approach (as compared to the Bézier curves approach) since an errant line needs to be erased and redrawn in full. Researchers should also be mindful to draw at a high resolution (i.e., on a large digital "canvas") to ensure that stylus tracing illustrations can be displayed at larger sizes without obvious pixelation (e.g., for conference posters or for large format presentations).

Software options for stylus tracing include subscription-based software such as Adobe Photoshop, and Adobe Illustrator, and single payment options, such as Procreate. Software with a basic free and more advanced paid versions, also referred to as "freemium" software, includes applications such as iBis Paint, Adobe Fresco, and

Sketchbook. In our experience, the free versions of these applications are sufficient for the type of illustrations discussed in this paper.

b. Bézier curves

For the generation of vector-based line illustrations such as those shown in Figs. 1(h), 2(h), 3, 4, and 5(a), we made use of a digital graphics tool—often called the *pen tool* in most software—to trace over selected frames of video. The pen tool generates strings of connected points (referred to as Bézier curves or paths) where each point contains the mathematical information of the smooth curve on either side of itself. An illustration created through this approach is, thus, comprised of a collection of paths. The creation of these lines is done by clicking and dragging at each point, releasing when the curve of the outline follows the shape of the object in the image being traced. This approach can have a steeper learning curve because of the need to learn the technique of drawing with Bézier curves, yet it has the advantage of creating vector-based illustrations that are in some ways easier to edit—since the points can be continually repositioned, the curves can be continually redefined—and can be made to clearly display at any size.

Software options for Bézier curves include most of the ones mentioned in the previous section, as well as the subscription-based CorelDRAW, as well as freely available Vectr, and the open-source Inkscape.

3. Vector vs raster

Vector-based images are images made using vector graphics rather than raster, or bitmap, graphics. In vector-based images, the lines, shapes, and colors are defined by the mathematical relationship between points in a 2D space rather than as pixel values. Both approaches mentioned above work with either vector or raster software, but creating illustrations within vector-based editors can afford a researcher two major advantages over raster-based approaches.

First, researchers using a vector-based approach can produce high-resolution illustrations at any scale. By nature of the form in which the information in a vector-based image is processed—as mathematical curves and values associated with coordinates in a plane—vector-based illustrations do not take on any sort of defined resolution until they have been converted to a raster version for printing or embedding in documents. In the same way that the idea of the plot of $y = x$ does not have a resolution until it is portrayed on a screen or printed out, vector-based images are not forced to have any resolution until they are saved as a raster-based file format such as JPEG, TIFF, or PNG (vector-based file formats such as SVG also exist but are generally less accepted by publishers than raster-based ones). This is not only a practical benefit to the researcher while they format manuscripts for printing or online publication but can also benefit the clarity of the illustration as compared to potentially low-resolution pictures or frames of video (as discussed in Sec. III B).

Second, researchers using a vector-based approach can precisely edit the details of an illustration in a manner that is efficient and reversible. Since the information of a vector-based illustration is processed mathematically, the vertices and edges of shapes can be repeatedly moved and shaped. If a portion of an outline is drawn incorrectly, for example, it need not be fully redrawn in order to correct the error. The illustrating researcher can simply alter the erroneous segment of the outline in isolation. Especially as a researcher generates various iterations of an illustration, the vector-based approach can give them precise control over the digital image without all or some of the illustration having to be redrawn each time something needs updating.

4. General tips for getting started

Throughout our article, we discussed many of the things you might want to consider when creating illustrations of your own in PER. In general, you should aim to depict the objects/environment/participants that you considered relevant for your analysis and, accordingly, you should exclude the things that were not part of your focus. However, there are several common exceptions to this rule that are worthwhile considering as you craft an illustration. First, objects irrelevant to your analyses may partially obscure parts of other, relevant objects that you wish to depict and, so, can be included in the final depiction. For example, in Fig. 8, there is a table in the foreground of the illustration that was not germane to our analysis, yet we included it in our depiction of the data to avoid having to construct the lower half of the seated participant or crop the illustration at the participant's waist. Second, there are often smaller, technically irrelevant details in an illustration that you may want to include for purposes of readability. For example, you may decide to draw lines to indicate a

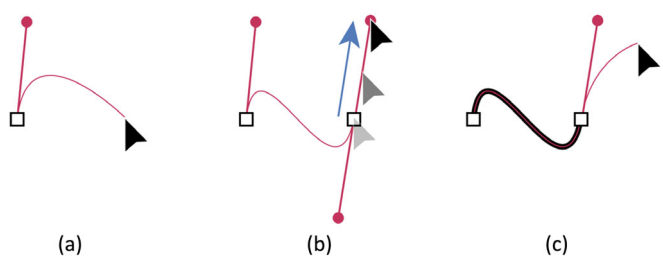

FIG. 8. A stepwise demonstration of how the so-called pen tool works to make Bézier curves. The Bézier curve "handles" are shown with straight lines ending with circles. These handles show the angle and "strength" of the tangent line for the thicker line at that point. (a) A line is previewed from a first point (the left square) toward a second point, with the mouse hovering (not clicked); (b) the mouse is clicked and dragged up and to the right, which causes the drawn line to curve upward into the second point; and (c) the mouse is released, committing the s-curved line between the first two points and starting the process all over.

participant's hair or you may choose to illustrate where the seams of the walls or floor were to better depict the orientation of the camera with respect to the space. Remember that throughout the stages of drafting an illustration, a guiding principle is to make the details of the scene easily interpretable (i.e., well defined against the background and self-explanatory). Thus, there is a balance to be struck on a case-by-case basis for the level of detail within a given illustration. Consider toggling to and from a version of the illustration where some of the drawn details are omitted to better determine how much those details enhance readability. As with academic writing, it can also help to catch any incorrect inclusions or omissions by zooming out or taking a break and returning to the illustration later.

We also recommend paying close attention to how you can visually represent the motions (i.e., gestures or changes in body position) and speech of the participants. As we have discussed in the article, a key advantage of illustrations is their flexibility to depict dynamic elements such as the movement of objects in the video. Additionally, readers may find it easier to interpret a scene if you place the coincident speech of the participants directly alongside the illustration as speech bubbles (see the bottom half of Fig. 3). For creating these annotations, most software options allow for the easy creation of arrows and speech bubbles (with the creation of rounded rectangles and a text box).

The "weight" (thickness) of your lines can also play a large role in how professional an image looks. An easy hierarchy can be made between the most important lines and least important lines with thicker and thinner line weights, respectively. In our work, we generally give the outermost line for each participant a thicker weight and the inner details of the participants a thinner weight to make the separation between multiple participants more readable. For those utilizing vector-based software, the weight of lines can be edited after the points are drawn. A similar effect can be achieved in raster-based approaches by altering the line weight (via the size of the digital pencil or brush) as you draw different elements in the scene. It can also be desirable to have the ends of your lines taper down to a thinner weight for a more naturally drawn look (as was done in Ref. [2], for example). Most vector-based software options allow for the tapering of lines after the curves have been drawn, while this effect can often be achieved automatically for many stylus tracing software options.

When drawing faces, it can be especially tricky to capture a likeness or expression with linework. Unfortunately, there is no turnkey solution to this. As mentioned above, try to toggle the number of details drawn on the faces of your participants. Usually, wrinkles around the mouth or eyes make a face read as older, so unless you are intentionally attempting to depict an older-aged participant, we recommend opting for fewer lines—draw simple versions of the eyes, nose, ears, and mouth first (often there is no need for lips, teeth, eyelashes, or nostrils) and see if this makes the expression readable already.

Adding color to images can be done perhaps most easily by using the "fill" tool, which fills in completed or closed shapes in the drawing with a selected color. Adding color can help highlight important features or represent different skin tones. Using a skin tone palette can be a useful aid here (see Ref. [67], for example). As discussed in the paper, we recommend that the researchers check with the portrayed participants if they agree with the way they were represented.

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